

SUDHIR KUMAR SINGH

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Seasoned software generalist building and scaling complex platforms, seeking to lead, deliver, and shape the future.

WORK EXPERIENCE

Walmart Inc

Senior Software Engineer (Eligibility team)

Sunnyvale, CA

Nov 2023 - Oct 2025

- Built high-throughput microservices responsible for deriving eligibility and restrictions for offers, items, and products at the offer, store, and fulfillment-center levels for global markets. (**Java17, WebFlux, Kafka, k8s**)
- Developed core eligibility infrastructure (including Cassandra integration and configurations) to support multitenant, store-level eligibility across Walmart International markets. (**Java17, WebFlux, Cassandra**)
- Architected backend enhancements in partnership with engineering, data science, and product teams to enable unified 1P/3P shipping workflows, improving delivery promise accuracy and reducing late deliveries by ~15%.
- Actively contributed to architectural and design discussions, code and design reviews, mentoring junior engineers, and driving improvements in monitoring, availability, operational support, and overall system reliability.

Strateos Inc

Software Engineer III

Menlo Park, CA

Jul 2021 - Jun 2023

- Designed and deployed Asset-management service components and UI (cloud lab) to remove tedious manual orchestration and management (Ansible scripts) of labs, workcells, and devices. (**Java17, Spring Boot, React**)
- Developed components using an event-driven architecture to compile a scientific intent into device commands; coordinating scheduling, execution, and reporting to reduce scientist time by ~90%. (**Scala, Akka, RabbitMQ**)
- Led a 2 developer effort and drove the implementation of an automated colony-picking system using a robotic arm and computer vision, achieving ~10× higher throughput than manual inspection. (**Python, OpenCV, CNN**)
- Owned end-to-end delivery of multiple product feature epics, from design through production deployment, collaborating with scientists, product, and platform teams.

Center On Access Technology, RIT

ML Software Engineer - (Part-time)

Rochester, NY

Jan 2021 - May 2021

- Built language and error models with ASR (Speech-To-Text) data to improve hybrid captioning for a hard-of-hearing class, and developed UI and backend services to store and serve ML models. (**Python, Flask, React**)
- Experimented and evaluated many NLP models, including transformer-based architectures in Hugging Face, to improve captioning accuracy on ASR datasets. (**PyTorch, BERT**)

Philips India Limited

Senior Software Engineer

Bangalore, India

Feb 2015 - Jul 2018

- Developed reliable REST APIs for the HealthSuite Digital Platform (HSDP) to enable secure exchange of patient health information between internal and external systems using HL7 and FHIR standards.
- Oversaw a team of 8 developers in the absence of a team architect and led the implementation of the Oncology integration project across two major platform releases.

CliniSys Solutions Limited

Software Engineer

Bangalore, India

Jun 2011 - Feb 2015

- Owned end-to-end product lifecycles with a team of 4 developers, covering requirements, design, development, deployment, and client interactions, while delivering 3 parallel product releases with **zero** client-reported issues.
- Developed product features across laboratory and pathology desktop and web solutions, including patient registration, results management, reporting, and HL7 message exchange for inpatient and outpatient workflows.

SKILLS (WORKED WITH TECH)

AI	: Sklearn, PyTorch, LLMs, Fine-Tuning, Agentic AI, MCP, A2A, Cursor, Claude code
Programming	: Python, Flask, FastAPI, Java, Bash, React, Spring Boot, WebFlux
Database/Tools	: VectorDB, Postgres, MongoDB, Cassandra, Git, Github, Gradle, Maven, Jenkins
DevOps/Infra	: AWS, Linux, Docker, Kubernetes, ArgoCD, Kafka, ELK, Splunk, Prometheus, Grafana
Others	: Jira, Confluence, Agile, REST, OpenAPI, Microservices, HIPAA

EDUCATION

Rochester Institute of Technology

MS Computer Science, GPA: 3.38

Rochester, NY

Aug 2018 - May 2021

- Classifying similar language varieties** (*Python, scikit-learn, NLP, NLTK, NumPy*) May 2019
- Developed ML models to classify similar languages and language varieties using the DSL dataset, leveraging TF-IDF-based character and word n-gram features with Naive Bayes and SVM models, achieving $\sim 88\%$ accuracy.
- CustomerSupport-FineTuned-Model** (*Mistral-7B, QLoRA, PEFT, Google Colab*) Feb 2026
- Fine-tuned an LLM for customer support auto-resolution to generate policy-aware, context-grounded responses by applying QLoRA on a 4-bit quantized Mistral-7B model trained on structured instruction datasets.
- HuggingFace Knowledge Assistant** (*Python, FastAPI, LangChain, FAISS, RAG*) Mar 2026
- Developed a RAG-based chatbot to provide accurate, context-aware answers by ingesting and embedding Hugging Face knowledge sources, leveraging semantic retrieval and LLM-based generation over indexed knowledge sources.
- miniAgent** (*Python, FastAPI, Asyncio, LangChain, Google ADK, CrewAI, LiteLLM*) Mar 2026 – Present
- Developing a lightweight, provider-agnostic orchestration layer for autonomous AI agents to eliminate vendor lock-in, enabling seamless backend transitions across multiple model providers, supporting both linear pipelines and complex graph workflows with conditional routing and asynchronous tool execution.